Consolidated Approach for Reporting Indicators of food security using remote data collection (remote-CARI) - Interim note[1]

Disclaimer: this guidance note is for limited use only until proper testing and face to face comparisons are possible. A final rCARI methodology and guidance will be developed shortly. This guidance is an internal document for the use of WFP staff only.

Introduction

In situations where access to affected populations is limited or non-existent, such as in case of conflict or epidemics, face to face interviews are not possible. In these circumstances, regular assessments are often replaced or complemented by remote data collection through call centres, Computerized Assisted Telephone Interviews (CATI) or web-based surveys. Remote surveys are short in length and do not include all the standard food security indicators that are normally used to assess food security and estimate the number of people in need. This note aims to provide a methodology to calculate CARI using proxy indicators currently collected through remote surveys. Examples of analysis outputs, syntaxes and data collection modules are also provided. The note is based on the best practices up to date but will be further reviewed and updated as further learnings will become available.

What is the remote-CARI?

Remote-CARI or rCARI is an approach used to aggregate different food security indicators into one index to report on population's overall food security status using remote surveys. This composite indicator is used to determine the number of food insecure people when data from regular assessments are not available due to access issues.

Other purposes of the rCARI include the monitoring of the food security situation over time and for profiling and classifying the food insecure to inform programme responses and targeting decisions.

Similar to the original CARI, r-CARI assesses two dimensions:

1) The **current status** of households' food consumption (assessed based on food consumption patterns)
2) The **current coping capacity** of households to meet future needs (assessed based on economic vulnerability and adoption of livelihood coping strategies)

What are the indicators collected through remote surveys?

Regular CARI assesses the current status in the consumption pattern using food consumption score and the coping capacity through livelihood coping strategies and food expenditure share (the latter being a proxy for economic vulnerability). Remote surveys have some limitations in collecting several of these indicators, due to length and complexity.

1) mVAM usually collects the following indicators that are relevant for rCARI indicator computation:
    1. Food consumption score
    2. Consumption based coping strategy (rCSI)
    3. Livelihood coping strategy (usually shorter version of the original)

[1] The remote CARI guidance is developed by WFP Needs Assessment and Targeting Unit (RAM) in close collaboration with significant inputs from the regional VAM units.

4. Income sources
5. Changes in income

2) Web-surveys can collect the following indicators:
   1. Consumption based coping strategy (modified version)
   2. Number of meals
   3. Worry about running out of food
   4. Livelihood coping strategy (short version)
   5. Income sources
   6. Change in income

The below table includes CARI indicators and some of the proxy indicators that could be used to calculate remote-CARI and its recommended thresholds.

Domain			Indicator	Food Secure (1)	Marginally Food Secure (2)	Moderately Insecure (3)	Severely Insecure (4)
Current Status	Food Consumption	**Preferred option**	Food consumption score	Acceptable	Acceptable consumption + using 'severe' food coping [2]	Borderline	Poor
		Alternative if preferred option not available (web survey)	Number of meals consumed	3 meals	2 meals	1 meal	0 meal
			Worry about food availability/food coping	Not worried	Worried	Worried + skipping meals	Worried + skipping day/night without eating
Current Coping Capacity	Economic vulnerability	**Only available option** to measure economic vulnerability remotely as expenditure cannot be collected *(classification needs to be contextualized)*	Main income source and change in income	**Regular employment** (formal labour or self-employed) – no change/ no decrease	**Regular employment** but reduced income or **informal labour/ remittances,** no change/ no decrease	**Informal labour /remittances** but reduced income	**No income**, dependent on assistance or support *Or informal labour with complete loss of income*
	Livelihood coping	**Preferred option** *(classification of strategies needs to be contextualized)*	Livelihood-based coping strategy categories	Neutral	Stress strategies	Crisis strategies	Emergency strategies (or do not have the capacity to do anything)
		Alternative if preferred option not available	Livelihood-based coping strategy number strategies (short version)	No strategies	1 strategy	2 strategies	3 strategies (or do not have the capacity to do anything)

Guidance on the prioritization of use of indicators and aggregation of remote-CARI:

1) Current food security status (50% of total score):
   a. Preferred option (recommended for CATI):
      - ✓ Two indicators (food consumption score and reduced coping strategy) will be used to classify households:
        - HH with acceptable consumption will take a score of 1,
        - HH with borderline consumption will take score of 3,
        - HH with poor consumption will take score of 4.
      - ✓ Reduced coping strategy index will be used to classify HH as 2 if the HH has acceptable consumption but has been using severe food coping strategies[2].
   b. Alternative (recommended for Web-surveys):

[2] Severe food coping refers to using any of the following strategies over the past week at least one time: reduced number of meals, reduce meals portion, reduce consumption by adults in order for children to eat. Further testing is required for the definition of severe coping.

- ✓ Three indicators will be used to classify HH into 4 categories (number of meals, worry about food availability and reduced coping strategies)
- ✓ Number of meals consumed in the day prior to the survey will be used to classify HH into 4 categories as mentioned in the table
- ✓ Worry about availability of food in addition to the use of food-based coping strategies (short version of the original module) will be used also to classify the HH into 4 categories as mentioned in the table
- ✓ The alternative methodology depends on perception-based indicator and measured at individual level. To reduce biases and strengthen results, the average value of the two classifications will be taken and rounded.

2) Economic vulnerability (25% of total score)
   a. Only available option:
      - ✓ Two indicators will be used in the classification of households (income sources and change in income)[3]
        - HH will be classified based on current income sources where HH with income coming from regular employment will be given a score of 1,
        - if the regular income source is accompanied by reduced income or the income is mainly from remittances or informal labour with no change or no decrease in income, the household will be given score of 2,
        - if the HH income source is informal labour and have decreased income[4], then the HH will be given a score of 3,
        - if the HH has no income, loss all income or mainly depends on assistance then it will be given a score of 4.

3) Livelihood coping (25% of total score)
   a. Preferred option:
      - ✓ Livelihood coping strategies (Full module with 10 strategies; 4 stress, 3 crisis, 3 emergency).
   b. Alternative option
      - ✓ Livelihood coping strategies are asked in three questions (one combines any of the listed stress strategies, and one for any of the listed crisis strategies and one for any of the listed emergency strategies)
   c. Alternative option (least recommended):
      - ✓ If three coping strategies were used, HH will be classified based on the number of coping strategies used as per the table
      - ✓ Households has no coping capacities would need to be downgraded (to be tested).

The compositive score is comprised of 50% of the current food consumption domain, 25% of economic vulnerability domain, and 25% of the livelihood coping domain.

rCARI versus IPC/CH analyses

The integrated food security phase classification (IPC) and the Cadre Harmonise (CH) analyses are among the main users of food security indicators collected remotely. If IPC/CH is taken place in a given country, WFP recommends using IPC/CH to report on the number of food insecure and their geographical distribution. rCARI should be used for profiling of food insecure population.

[3] The aggregation of the income sources should be contextualized
[4] Change in income could be contextualized to include households that experienced more than 25% or 50% reduction in income

In the absence of IPC or when numbers are outdated, the rCARI can be used to estimate the number of food insecure at national level, at sub-national level or for defined geographic areas or socio-economic groups.

Annexes to this guidance note:

Annex 1 provides an example on the results of a pilot-test that was conducted using data recently collected in Lebanon using web-survey **(note these are just test results and have not yet been reported)**. Also, a generic syntax for web-survey analysis is provided.

Annex 2 provides an example on the results of a pilot-test that was conducted using data recently collected in Uganda using mVAM **(note these are just test results and have not yet been reported)**. Also, a generic syntax for mVAM analysis is provided.

Annexes 3 and 4 provide an overview of questions and modules for web and phone surveys to collect all rCARI relevant questions.

Annex I: Pilot-test example - Lebanon web-survey

Remote CARI console for (Lebanese population)

Domain			Indicator	Food Secure (1)	Marginally Food Secure (2)	Moderately Insecure (3)	Severely Insecure (4)
Current status	Food Consumption	Alternative if preferred option not available (web survey)	Number of meals consumed	15.8%	54.9%	25.6%	3.7%
			Worry about food availability/food coping	49.5%	35.6%	11.0%	3.9%
Current Coping Capacity	Economic vulnerability	**Only available option** to measure economic vulnerability remotely as expenditure cannot be collected remotely *(classification needs to be contextualized)*	Main income source and change in income	19.6%	44.9%	20.8%	14.7%
	Livelihood coping	**Preferred option** *(classification of strategies needs to be contextualized)*	Livelihood-based coping strategy categories	31.1%	35.9%	14.7%	18.3%
r-CARI				24.4%	49.8%	23.6%	2.2%

		r-CARI			
		Food secure (1)	**Marginally food secure (2)**	**Moderately food insecure (3)**	**Severely food insecure (4)**
		Row N %	Row N %	Row N %	Row N %
Resident status	Lebanese	24.4%	49.8%	23.6%	2.2%
	Syrian refugees	8.0%	42.9%	43.2%	5.9%
	Palestinian refugees	17.8%	41.5%	37.8%	2.9%

		Resident status		
		Lebanese	Syrian refugees	Palestinian refugees
		Column N %	Column N %	Column N %
Number of meals	Three	15.8%	6.2%	8.0%
	Two	54.9%	50.2%	62.5%
	One	25.6%	34.7%	23.9%
	One meal & going day/night without eating	3.7%	8.8%	5.7%

		Resident status		
		Lebanese	Syrian refugees	Palestinian refugees
		Column N %	Column N %	Column N %
Worried about not having enough food to eat	Not worried	49.5%	25.3%	36.9%
	Worried	35.6%	44.9%	43.9%
	Worried & skipping meals	11.0%	19.3%	12.0%
	Worried & going day/night without eating	3.8%	10.4%	7.1%

		Resident status		
		Lebanese	Syrian refugees	Palestinian refugees
		Column N %	Column N %	Column N %
Food-based coping	No difficulties eating enough	32.3%	17.2%	20.4%
	Ate less expensive food	43.2%	45.6%	52.6%
	Skipped meals	13.9%	21.3%	15.3%
	Went for a day/night without eating	4.9%	11.2%	7.9%

	Increased my food intake	5.7%	4.6%	3.8%

		Resident status		
		Lebanese	Syrian refugees	Palestinian refugees
		Column N %	Column N %	Column N %
Household income	Regular employment (formal labour or self-employed) – no change or increase due to COVID	19.6%	9.1%	10.0%
	Regular employment but reduced income or informal labour/ remittances no change/ increase due to COVID	44.9%	34.9%	35.4%
	Informal labour /remittances but reduced income due to COVID	20.8%	38.7%	30.7%
	No income, dependent on assistance or support	14.7%	17.3%	23.9%

		Resident status		
		Lebanese	Syrian refugees	Palestinian refugees
		Column N %	Column N %	Column N %
Summary of asset depletion	No coping strategies (no need)	31.1%	21.5%	26.3%
	Stress coping strategies	35.9%	33.6%	40.3%
	Crisis coping strategies	14.7%	8.7%	9.8%
	Emergencies coping strategies or no capacity	18.4%	36.2%	23.6%

Syntax

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******************************WEB METHOD*************************

/* This syntax is an example. Due to small differences in the questionnaires, it needs to be validated and adjusted for the country dataset.

************Food Consumption************

*****Meals ******

********define variables:

RESPMeals=1                 'One or none'
RESPMeals=2                 'Two'
RESPMeals=3                 'Three or more'
RESPPercFoodSec_1M = 4      'Went at least one whole day and night without eating'

IF (RESPMeals = 3) CARI_Meals = 1.
IF (RESPMeals = 2) CARI_Meals = 2.
IF (RESPMeals = 1) CARI_Meals = 3.
IF (RESPMeals =0) CARI_Meals = 4.
IF (RESPMeals =1 & RESPPercFoodSec_1M = 4) CARI_Meals = 4.  /* There is no option for zero meals, it is proxied as stated here.

VALUE LABELS CARI_Meals
1 'Three meals'
2 'Two meals'
3 'One meal'
4 'No meals'.
EXECUTE.

FREQUENCIES CARI_Meals.

*****Worry+Coping*****
********------define variables:
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RESPFoodWorry_YN_1M=1        'In the past 7 days, felt worried about not having enough food to eat?'
RESPFoodWorry_YN_1M=2        'In the past 7 days, did not feel worried about not having enough food to eat?'

RESPPercFoodSec_1M=3         'Skipped meals or ate less than usual'
RESPPercFoodSec_1M=4         'Went at least one whole day and night without eating'

IF (RESPFoodWorry_YN_1M =2) CARI_Worry = 1.
IF (RESPFoodWorry_YN_1M =1) CARI_Worry = 2.
IF (RESPFoodWorry_YN_1M =1 & RESPPercFoodSec_1M =3) CARI_Worry = 3.
IF (RESPFoodWorry_YN_1M =1 & RESPPercFoodSec_1M = 4) CARI_Worry = 4.

VALUE LABELS CARI_Worry
1 'Not worried'
2 'Worried'
3 'Worried & skipping meals'
4 'Worried & going day/night without eating'.
EXECUTE.

FREQUENCIES CARI_Worry.

************Economic vulnerability************
********------define variables:
HHIncFirst =1          'Salaried work'
HHIncFirst =2          'Daily casual labour'
HHIncFirst =3          'Own business trade'
HHIncFirst =4          'Petty trade selling on streets'
HHIncFirst =5          'Farming livestock production'
HHIncFirst =6          'Remittances from abroad'
HHIncFirst =7          'Support from family friends'
HHIncFirst =8          'Government assistance'
HHIncFirst =9          'Support from UN/NGO charity'
HHIncFirst =10         'Begging/scavenging'
HHIncFirst =11         'Living off of debt'
HHIncFirst =12         'Other'


HHIncCh=1              'Increased'
HHIncCh=2              'No change'
HHIncCh=3              'Reduced by less than 25%'
HHIncCh=4              'Reduced by more than 25% and less than 50%'
HHIncCh=5              'Reduced by more than 50%'
HHIncCh=6              'Alternative income resources'          /* Alternative income resources for the same level of income.

IF (HHIncFirst =1 | HHIncFirst = 3 | HHIncFirst=5) & (HHIncCh=1 | HHIncCh=2 | HHIncCh=6) CARI_Inc =1.
IF (HHIncFirst =1 | HHIncFirst =3 | HHIncFirst=5) & (HHIncCh=3| HHIncCh=4| HHIncCh=5) CARI_Inc =2.
IF (HHIncFirst =2 | HHIncFirst=4 | HHIncFirst=11 | HHIncFirst=6) & (HHIncCh=1 | HHIncCh=2 | HHIncCh=6) CARI_Inc =2.
IF ((HHIncFirst =2 | HHIncFirst=4 | HHIncFirst=11 | HHIncFirst=6) & (HHIncCh=3| HHIncCh=4| HHIncCh=5)) CARI_Inc =3.
IF (HHIncFirst =7 | HHIncFirst =8 | HHIncFirst=9 | HHIncFirst=10) CARI_Inc =4.
IF (HHIncFirst =12) CARI_Inc=5.   */ If no further information is available, other needs to be set to missing. We suggest either case wise
deletion, or to leave economic vulnerability missing, but reweight the other indicators for building the index.

RECODE CARI_Inc (5=SYSMIS) INTO CARI_Inc.
EXECUTE.

VALUE LABELS CARI_Inc
1 ' Regular employment (formal labour or self-employed) – no change or increase due to COVID'
2 ' Regular employment but reduced income or informal labour/ remittances no change/ increase due to COVID
3 ' Informal labour /remittances but reduced income due to COVID'
4 ' No income, dependent on assistance or support.
EXECUTE.


FREQUENCIES CARI_Inc.

************LIVELIHOOD COPING************
*stress strategies
********------define variables:
HHCoping_1M_spend_savings=3          'Spent savings'
HHCoping_1M_spent_less_on_food=3     'Spent less on food'
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HHCoping_1M_ask_family_friends_for_help=3          'Asked family/friends for help '
HHCoping_1M_borrowing_money_using_credit=3         'Borrowed money or bought on credit'

DO IF (HHCoping_1M_spend_savings=3 OR HHCoping_1M_spent_less_on_food=3 OR HHCoping_1M_ask_family_friends_for_help=3 OR
HHCoping_1M_borrowing_money_using_credit=3).

COMPUTE Stress_coping = 1.
ELSE.
COMPUTE Stress_coping = 0.
END IF.
EXECUTE.

*crisis strategies
********------define variables:
HHCoping_1M_selling_productive_assets=3            'Sold productive assets'
HHCoping_1M_spent_less_on_health=3                 'Spent less on health'
HHCoping_1M_spent_less_on_education=3              'Spent less on education'

DO IF (HHCoping_1M_selling_productive_assets=3 OR HHCoping_1M_spent_less_on_health=3 OR
HHCoping_1M_spent_less_on_education=3).
COMPUTE Crisis_coping = 1.
ELSE.
COMPUTE Crisis_coping = 0.
END IF.
EXECUTE.

*emergency strategies
********------define variables:
HHCoping_1M_send_children_to_work=3                'Sent children to work'
HHCoping_1M_begging_scavenging=3                   'Begged/scavenged'
HHCoping_1M_sell_house_or_land=3                   'Sold house or land'

DO IF (HHCoping_1M_send_children_to_work=3 OR HHCoping_1M_begging_scavenging=3 OR HHCoping_1M_sell_house_or_land=3 OR
HHCoping_1M_spend_savings=2 OR HHCoping_1M_spent_less_on_food=2 OR HHCoping_1M_ask_family_friends_for_help=2 OR
HHCoping_1M_borrowing_money_using_credit=2 OR HHCoping_1M_selling_productive_assets=2  OR
HHCoping_1M_spent_less_on_health=2      OR
HHCoping_1M_spent_less_on_education=2 OR HHCoping_1M_send_children_to_work=2 OR HHCoping_1M_begging_scavenging=2 OR
HHCoping_1M_sell_house_or_land=2).

COMPUTE Emergency_coping = 1.
ELSE.
COMPUTE Emergency_coping = 0.
END IF.
EXECUTE.

VARIABLE LABELS Stress_coping 'did HH engage in stress coping strategies?'.
VARIABLE LABELS Crisis_coping 'did HH engage in crisis coping strategies?'.
VARIABLE LABELS Emergency_coping 'did HH engage in emergency coping strategies?'.
EXECUTE.

RECODE Stress_coping (0=0) (1=2).
RECODE Crisis_coping (0=0) (1=3).
RECODE Emergency_coping (0=0) (1=4).
COMPUTE Max_coping_behaviour=MAX (Stress_coping, Crisis_coping, Emergency_coping).
RECODE Max_coping_behaviour (0=1).

VALUE LABELS Max_coping_behaviour
1 'No coping strategies'
2 'Stress coping strategies '
3 'Crisis coping strategies '
4 'Emergencies coping strategies'.
EXECUTE.

VARIABLE LABELS Max_coping_behaviour 'Summary of asset depletion'.

FREQUENCIES Max_coping_behaviour.


**** rCARI
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COMPUTE rCARI_mean=SUM ((CARI_Meals*0.25),(CARI_Worry*0.25), (CARI_Inc*0.25), (Max_coping_behaviour*0.25)).
COMPUTE rCARI=RND (rCARI_mean).
EXECUTE.
VALUE LABELS rCARI 1 "Food secure" 2 "Marginally food secure" 3"Moderately food insecure" 4 "Severely food insecure".
```

Annex II: Pilot-test example – Uganda mVAM

Remote CARI console for (Ugandan nationals)

Domain			Indicator	Food Secure (1)	Marginally Food Secure (2)	Moderately Insecure (3)	Severely Insecure (4)
Current status	Food Consumption	**Preferred option**	Food consumption score	31.1%	38.3%	26.4%	4.2%
Current Coping Capacity	Economic vulnerability	**Only available option** to measure economic vulnerability remotely as expenditure cannot be collected remotely *(classification needs to be contextualized)*	Main income source and change in income	6.3%	70.6%	19.1%	4.1%
	Livelihood coping	**Preferred option** *(classification of strategies needs to be contextualized)*	Livelihood-based coping strategy categories	5.7%	28.0%	41.5%	24.8%
r-CARI				4.4%	49.6%	43.2%	2.8%

		r-CARI			
		Food secure (1)	**Marginally food secure (2)**	**Moderately food insecure (3)**	**Severely food insecure (4)**
		Row N %	Row N %	Row N %	Row N %
Resident status	Ugandan Nationals	4.43%	49.62%	43.18%	2.78%
	Refugees in settlement	1.25%	17.26%	60.11%	21.38%
	Refugees in Kampala	0.21%	12.63%	67.37%	19.79%

		Resident status		
		Ugandan Nationals	Refugees in settlement	Refugees in Kampala
		Column N %	Column N %	Column N %
Food Consumption Score	Acceptable	31.08%	4.52%	8.42%
	Acceptable consumption but using severe coping	38.27%	21.04%	44.63%
	Borderline	26.35%	47.86%	33.47%
	Poor	4.29%	26.58%	13.47%

		Resident status		
		Ugandan Nationals	Refugees in settlement	Refugees in Kampala
		Column N %	Column N %	Column N %
Household income	Regular employment (formal labour or self-employed) – no change or increase due to COVID	6.26%	3.42%	0.63%
	Regular employment but reduced income or informal labour/ remittances no change/ increase due to COVID	70.56%	28.13%	23.16%
	Informal labour /remittances but reduced income due to COVID	19.10%	29.97%	46.53%
	No income, dependent on assistance or support	4.07%	38.49%	29.68%

		Resident status		
		Ugandan Nationals	Refugees in settlement	Refugees in Kampala
		Column N %	Column N %	Column N %
Summary of asset depletion	No coping strategies (no need)	5.67%	23.70%	0.21%
	Stress coping strategies	27.99%	16.62%	2.11%
	Crisis coping strategies	41.50%	27.82%	25.89%
	Emergencies coping strategies or no capacity	24.85%	31.86%	71.79%

Syntax- CATI

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**************************************CATI METHOD**************************************
************FOOD CONSUMPTION SCORE (FCS) ************

COMPUTE FCS= SUM (FCSStap*2, FCSPulse*3, FCSDairy*4, FCSPr*4, FCSVeg*1, FCSFruit*1, FCSFat*0.5, FCSSugar*0.5).
EXECUTE.

RECODE FCS (Lowest thru 28.00=1) (28.5 thru 42.00=2) (42.5 thru Highest=3) INTO FCG.
EXECUTE.

****************************Recode the food consumption groups into the 4-point scale ********************************

RECODE FCG (1=4) (2=3) (3=1) INTO FCS_4pt.
Variable labels FCS_4pt '4pt_FCG'.
Execute.

FREQUENCIES VARIABLES=FCS_4pt.

**Define Variable Properties. **
*FCS_4pt.

VALUE LABELS FCS_4pt
1 'Acceptable'
2 'Acceptable consumption but using severe coping'
3 'Borderline'
4 'Poor'.
EXECUTE.

/*Severe food coping refers to using any of the following strategies over the past week at least one time:
/*reduced number of meals, reduce meals portion, reduce consumption by adults in order for children to eat

*********** REDUCED COPING STRATEGIES*********************

IF ((rCSIMealSize GE 1 ) | ( rCSIMealNb GE 1) | (rCSIMealAdult GE 1)) severe_food=1.
EXECUTE.

FREQUENCIES severe_food FCS_4pt.


DO IF (FCS_4pt =1 & severe_food=1).
RECODE FCS_4pt (1=2).
END IF.
FREQUENCIES FCS_4pt.

************ECONOMIC VULNERABILITY************

VALUE LABELS livelihood_cat
1 'Negative impact/reduced income'
2 'No impact/no change'
3 'Positive impact/increased income'.
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EXECUTE.

FREQUENCIES livelihood_cat.

VALUE LABELS HHIncFirst
1 'Food crop production/sales'
2 'Cash crop production/sales'
3 'Agricultural wage labour'
4 'Income derived from sale of livestock and/or animal products'
5 'Non-agricultural wage labour'
6 'Small business/petty trade'
7 'Wholesale or large food/non-food store'
8 'Sale of firewood/charcoal'
9 'Transportation/driver'
10 'Skilled daily work'
11 'Pension'
12 'Salary'
14 'Handicrafts'
15 'Begging/ scavenging'
16 'Gifts/donations from NGOs/Government/UN'
18 'Remittances'.
EXECUTE.

RECODE HHIncFirst (1=1) (2=1) (3=2)
(4=1) (5=2) (6=1) (7=1) (8=1) (9=1)
(10=2) (11=1) (12=1)
(14=2) (15=4) (16=4) (18=2) into HHIncFirst_Cat.
EXECUTE.

VALUE LABELS HHIncFirst_Cat
1 'Regular'
2 'Informal or remittances'
4 'Assistance or no income'.
EXECUTE.

IF (HHIncFirst_Cat=1 ) EconomicVulnerability=1.
IF (HHIncFirst_Cat=2 | (HHIncFirst_Cat=1 & livelihood_cat=1)) EconomicVulnerability=2.
IF ((HHIncFirst=18  | HHIncFirst_Cat=2) & livelihood_cat=1) EconomicVulnerability=3.
IF (HHIncFirst_Cat=4) EconomicVulnerability=4.
EXECUTE.

FREQUENCIES EconomicVulnerability.

VALUE LABELS EconomicVulnerability
1 'Regular income'
2 'Regular income decreased or informal income and remittances'
3 'Informal or remittance and decreased'
4 'No income or assistance'.
EXECUTE.

************LIVELIHOOD COPING************

/*LhCSIBorrowCash_cat
/*LhCSISaving_cat
/*LhCSIDomAsset_cat
/*LhCSIEatOut_cat
/*LhCSIProdAsset_cat
/*LhCSIOutSchool_cat
/*LhCSIHealth_cat
/*LhCSISellAni_cat
/*LhCSIResAsset_cat
/*LhCSIBegged_cat
/*LhCSISeed_cat
/*LhCSIIllegalAct_cat
/*LhCSIMigrationPart_cat
/*LhCSILastfemAni_cat
/*LhCSIMoveOut_cat
/* No it was not necessary=1 – No exhausted=2 – Yes=3 - Not Applicable=4
```

```
/*stress strategies
/*****define variables:
/*LhCSIBorrowCash_cat ‘Borrowed money’
/*LhCSISaving_cat 'Spent savings'
/*LhCSIDomAsset_cat 'Sold household assets'
/*LhCSIEatOut_cat 'Sent household members to eat elsewhere'


DO IF (LhCSIBorrowCash_cat=3 OR LhCSISaving_cat=3 OR LhCSIDomAsset_cat=3 OR LhCSIEatOut_cat=3).

COMPUTE Stress_coping = 1.
ELSE.
COMPUTE Stress_coping = 0.
END IF.
EXECUTE.

*crisis strategies
*****define variables:
/*LhCSIProdAsset_cat 'Sold productive assets'
/*LhCSIOutSchool_cat 'Withdrew children'
/*LhCSIHealth_cat 'Reduce non-food health expenditure'

DO IF (LhCSIProdAsset_cat=3 OR LhCSIOutSchool_cat=3 OR LhCSIHealth_cat=3).
COMPUTE Crisis_coping = 1.
ELSE.
COMPUTE Crisis_coping = 0.
END IF.
EXECUTE.

*emergency strategies
*****define variables:
/*LhCSIResAsset_cat 'Sold house or land'
/*LhCSIBegged_cat 'Begged'
/*LhCSIIllegalAct_cat 'Involved in risky/illegal work/behaviour'
/*LhCSIMoveOut_cat 'Had to move out'

DO IF (LhCSIResAsset_cat=3 OR LhCSIBegged_cat=3 OR LhCSIIllegalAct_cat=3 OR LhCSIMoveOut_cat=3 OR LhCSIBorrowCash_cat=2 OR LhCSISaving_cat=2 OR LhCSIDomAsset_cat=2 OR LhCSIEatOut_cat=2 OR LhCSIProdAsset_cat=2 OR LhCSIOutSchool_cat=2 OR LhCSIHealth_cat=2 OR LhCSIProdAsset_cat=2 OR LhCSIOutSchool_cat=2 OR LhCSIHealth_cat=2).
COMPUTE Emergency_coping = 1.
ELSE.
COMPUTE Emergency_coping = 0.
END IF.
EXECUTE.

VARIABLE LABELS Stress_coping 'did HH engage in stress coping strategies?'.
VARIABLE LABELS Crisis_coping 'did HH engage in crisis coping strategies?'.
VARIABLE LABELS Emergency_coping 'did HH engage in emergency coping strategies?'.
EXECUTE.

RECODE Stress_coping (0=0) (1=2).
RECODE Crisis_coping (0=0) (1=3).
RECODE Emergency_coping (0=0) (1=4).
COMPUTE Max_coping_behaviour=MAX (Stress_coping, Crisis_coping, Emergency_coping).
RECODE Max_coping_behaviour (0=1).

VALUE LABELS Max_coping_behaviour
1 'No coping strategies'
2 'Stress coping strategies '
3 'Crisis coping strategies '
4 'Emergencies coping strategies'.
EXECUTE.

Variable Labels Max_coping_behaviour 'Summary of asset depletion'.

**** rCARI
```

```
Compute Mean_coping_capacity = MEAN (Max_coping_behaviour, EconomicVulnerability).
Compute FS_class_unrounded = MEAN (FCS_4pt, Mean_coping_capacity).
Compute FS_final= RND (FS_class_unrounded).
value labels FS_final
1 "Food secure" 2 "Marginally food secure" 3"Moderately food insecure" 4 "Severely food insecure".
execute.
```

Annex III: Data collection module web survey

Food consumption

Worried about food (RESPFoodWorry)

In the past 7 days, how did you feel about your household access to food?

1. I felt worried not to have enough food
2. I did not feel worried about having enough food

Food coping strategies (short version, RESPPercFoodSec)

Which statement best reflects your food situation over the past 7 days?

1- Had no difficulties eating enough food
2- Ate less expensive or less preferred food
3- Skipped meals or ate less than usual
4- Went at least one whole day and night without eating

Economic vulnerability

Income sources (HHIncFirst)

What was your household's main income source over the past year? *(NEEDS TO BE CONTEXTUALIZED)*

1- Salaried work with regular income
2- Own business/trade
3- Petty trade/selling on streets
4- Farming/livestock production
5- Daily/casual labour
6- Remittances from abroad
7- Support from families and friends
8- Government assistance/social safety nets
9- Support from UN/NGO/charity
10- High risk activity (e.g. begging (asking strangers for food or money, scavenging)
11- Living off on debt
12- Other

Change of income source (RESPIncCh)

Has your source of income been affected (by the Corona Virus / COVID-19 pandemic)? *(ADJUST FOR APPROPRIATE RECALL PERIOD)*

1. No, there was no change
2. Yes, income increased
3. Yes, income decreased by at least 25%
4. Yes, income decreased by at least 50%
5. Yes, income decreased by at least 75%
6. Resorted to alternative income sources to maintain previous level

Livelihood coping

Livelihood coping strategy categories (preferred option, HHCoping_1M)

In the past 30 days, have you or anyone in your household engaged in any of the following activities because there were not enough resources to meet your food needs? Check all that apply (COs needs to adjust strategies following existing guidance)

1. Spend savings

2. Selling domestic assets (TV, cell phones, jewelry, etc.)
3. Selling productive assets (used for work purposes)
4. Ask family/friends for help
5. Borrowing money or buying on credit
6. Diversify income activities
7. Spent less on health
8. Spent less on education
9. Spent less on food
10. Send children to work
11. Begging/scavenging
12. Sell house or land
13. Other
14. Did not do anything, because there was no need → skip to next question
15. Did not have the capacity to do something → skip to next question
16. No additional strategy used (for those who selected at least one strategy)

Livelihood coping strategy count (alternative short version)

During the last 30 days, did you or anyone in your household **spent savings, sold non-productive assets or borrowed food** due to a lack of food or lack of money to buy food? (The CO should include the most frequent stress coping strategies in the body of the question based on the context)

1. No - because it was not necessary
2. No - because strategy already exhausted and cannot do it anymore
3. Yes

During the last 30 days, has your household **sold productive assets/goods, withdraw children from school or reduced spending on health/education** due to a lack of food or lack of money to buy food? (Examples: sell the sewing machine, female breeding animals if working in agriculture) (The CO should include the most frequent crisis coping strategies in the body of the question based on the context)

1. No - because it was not necessary
2. No - because strategy already exhausted and cannot do it anymore
3. Yes

During the last 30 days, has your household **send a family member to beg, sent children to work, or marry a daughter/sibling before age of 18** to meet the food needs? (The CO should include the most frequent crisis coping strategies in the body of the question based on the context)

1. No - because it was not necessary
2. No - because strategy already exhausted and cannot do it anymore
3. Yes

Annex IV: Data collection module CATI

Food Consumption

Food Consumption Score	
NoteFCS	Now I would like to ask you about the foods and drinks your household ate or drank in the last 7 days during the day and at night [breakfast, lunch, dinner, snacks]. 1)NEXT
FCSStap	How many days in the last 7 days did your household eat Cereals, grain, roots, tubers and matooke? (Matooke, Rice, bread, millet, maize, chapatti, sorghum,potato, yam, cassava, sweet potato, and / or other tubers) NEEDS TO BE CONTEXTUALIZED [OPERATOR: RECORD NUMBER OF DAYS 0 - 7]
FCSPulse	How many days in the last 7 days did your household eat Legumes / nuts? (ground nuts, peanuts, sim-sim, coconuts or other nuts, beans, cowpeas, lentils, soy, pigeon pea) [OPERATOR: RECORD NUMBER OF DAYS 0 - 7]
FCSDairy	How many days over the last 7 days, did members of your household eat or consume a substantial amount of milk (powdered or fresh) or other dairy products (such as yogurt, cheese, condensed milk, sour cream or others)? (Excluding butter/margarine)? [OPERATOR: RECORD NUMBER OF DAYS 0 - 7]
FCSPr	How many days over the last 7 days, did members of your household eat or consume: meat, fish and eggs (such as goat meat, beef, chicken, seafood, canned tuna, etc.) as a main dish, not as a condiment? [OPERATOR: RECORD NUMBER OF DAYS 0 - 7]
FCSVeg	How many days in the last 7 days did your household eat Vegetables(orange, green, other) ? [OPERATOR: RECORD NUMBER OF DAYS 0 - 7]
FCSFruit	How many days over the last 7 days, did members of your household eat or consume: fruits (such as banana, apple, lemon, mandarin, mango, papaya, etc.)? [OPERATOR: RECORD NUMBER OF DAYS 0 - 7]
FCSFat	How many days over the last 7 days, did members of your household eat or consume: oil/fat/butter (such as cooking oil, butter, margarine, other fat/oil)? [OPERATOR: RECORD NUMBER OF DAYS 0 - 7]
FCSSugar	How many days over the last 7 days, did members of your household eat or consume: sugar or sugar products (such as honey, jam, candies, biscuits, pastries, cakes, and/or other sugar products)? [OPERATOR: RECORD NUMBER OF DAYS 0 - 7]

Economic vulnerability

Income sources	
HHIncFirst	What is the main income source of your household at the moment? [OPERATOR: DO NOT READ OUT THE RESPONSE OPTION, SELECT THE RESPONSE OPTION THAT BEST FITS THE INFORMATION PROVIDED BY THE RESPONDENT. IF THERE ARE MULTIPLE REASONS, ASK THE RESPONDENT WHICH ONE IS THE MAIN REASON AND SELECT THAT RESPONSE OPTION.] 1)SALARIED WORK WITH REGULAR INCOME IN THE PUBLIC OR PRIVATE SECTOR 2)INFORMAL DAILY/CASUAL LABOUR 3)BUSINESS/SELF-EMPLOYED [FORMAL] 4)INFORMAL TRADE/STREET SALES 5)CLEANING/CARE WORK AT OTHER PEOPLE'S HOME 6)NO INCOME SOURCE 7)GOVERNMENT ASSISTANCE/SOCIAL PROTECTION PROGRAMMES/PENSIONS

	8)SUPPORT FROM UN/NGO/CHARITY 9)MIGRANT REMITTANCES OR HELP FROM FAMILY AND/OR FRIENDS

Income change (ADJUST FOR APPROPRIATE RECALL PERIOD)	
HHIncCh	Has your household income changed since the Corona Virus / COVID-19 outbreak? [OPERATOR: SELECT ONE RESPONSE OPTION.] 1)YES 2)NO
HHIncChType	How has your household income changed since the Corona Virus / COVID-19 outbreak? [OPERATOR: DO NOT READ OUT THE RESPONSE OPTION, SELECT THE RESPONSE OPTION THAT BEST FITS THE INFORMATION PROVIDED BY THE RESPONDENT. IF THERE ARE MULTIPLE REASONS, ASK THE RESPONDENT WHICH ONE IS THE MAIN REASON AND SELECT THAT RESPONSE OPTION.] 1)THE INCOME DECREASED 2)THE INCOME INCREASED

Livelihood coping

LCSI (long version) - *CHOOSE STRATEGIES BASED ON CONTEXT*		
LhCSI	During the past 30 days, did you or anyone in your household have to XXXXXXXXXXXXX due to a lack of food or a lack of money to buy food? [OPERATOR: CHOOSE ONLY ONE OPTION] 1)No - because it wasn't necessary 2)No - because strategy already exhausted and cannot use it anymore 3)Yes	
Stress strategies	<u>Stress (choose 4)</u> - Sold household assets/goods (radio, furniture, television, jewelry etc.) - Spent savings - Sold more animals (non-productive) than usual - Sent household members to eat elsewhere - Purchased food on credit or borrowed food - Borrowed money - Move children to less expensive school	LhCSIDomAsset LhCSISaving LhCSIAnimals LhCSIEatOut LhCSICrdtFood LhCSIBorrowCash LhCSILessSchool
Crisis strategies	<u>Crisis (choose 3)</u> - Sold productive assets or means of transport (sewing machine, wheelbarrow, bicycle, car, etc.) - Withdrew children from school - Reduced expenses on health (including drugs) and education - Harvested immature crops (e.g. green maize) - Consumed seed stocks that were to be saved for the next season - Decreased expenditures on fertilizer, pesticide, fodder, animal feed, veterinary care, etc - Taking on debts or informal loans to cover basic needs	LhCSIProdAsset LhCSIOutSchool LhCSIHealth LhCSIImmCrops LhCSISeed LhCSIAgriCare
Emergency strategies	<u>Emergency (choose 3)</u> - Sold house or land - Begged - Engaged in illegal income activities (theft, prostitution) - Sold last female animals - Entire household migrated - Scavenging	LhCSIResAsset LhCSIBegged LhCSIIllegalAct LhCSIFemAnimal LhCSIMigration

LCSI – short version *CHOOSE STRATEGIES BASED ON CONTEXT*	
NoteLCSI	Now I I will ask you about the use, in your home, of some of the actions that I will mention, during the last 30 days to cope with the lack of food or money to buy food.

	1)CONTINUE
LhCSIHealth	During the last 30 days, has your household reduced essential non-food expenditure such as education and health to meet its food needs? [OPERATOR: SELECT ONLY ONE OPTION] [IF THE RESPONDENT ANSWERS NO, PLEASE PROVE CAREFULLY THE REASON, LISTEN AND MARK THE CORRESPONDING OPTION] 1)No. Because this was not necessary 2)No. Because I have already carried out this activity during the last 12 months and I cannot continue to do so 3)Yes 4)Not Applicable
LhCSISaving	During the last 30 days, has your household spent savings to meet its food needs? [OPERATOR: SELECT ONLY ONE OPTION] [IF THE RESPONDENT ANSWERS NO, PLEASE PROVE CAREFULLY THE REASON, LISTEN AND MARK THE CORRESPONDING OPTION] 1)No. Because this was not necessary 2)No. Because I have already carried out this activity during the last 12 months and I cannot continue to do so 3)Yes 4)Not Applicable
LhCSIDomAsset	During the last 30 days, has your household sold productive assets/goods to meet food or other needs? (Examples: sell the sewing machine if the work is seamstress; sell maazadon, shovel, piocha, machete, female breeding animals if working in agriculture) [OPERATOR: SELECT ONLY ONE OPTION] [IF THE RESPONDENT ANSWERS NO, PLEASE PROVE CAREFULLY THE REASON, LISTEN AND MARK THE CORRESPONDING OPTION] 1)No. Because this was not necessary 2)No. Because I have already carried out this activity during the last 12 months and I cannot continue to do so 3)Yes 4)Not Applicable